INCH-POUND

MIL-M-28787/317A
AMENDMENT 1
16 December 1992

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

REGISTER, DIGITAL, KEY CODE PJB

This amendment forms a part of Military Specification MIL-M-28787/317A(EC), dated 27 April 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4 THROUGH 8

TABLE I: Delete and substitute new table I as printed on pages 4 through 15 of this amendment.

PAGE 9

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through SS shall be performed in any order with the following exceptions. Test conditions A, B, C, and D are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, and D where applicable, followed by the remaining tests in the sequence."

4.5.3: Delete and substitute the following:

"4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through RR shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests E through RR shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I."

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PAGE 20

TABLE V, Test conditions C1 through C22, Min column: Delete "-0.4" (one place) and substitute "-6.0" (one place).

TABLE V, Test conditions C1 through C22, Max column: Delete "-6.0" (one place) and substitute "-0.4" (one place).

PAGES 37 AND 38

TABLE XIV, Test conditions M1 through M22, Min column: Delete "0" (two places) and substitute "-200" (two places).

TABLE XIV, Test conditions M1 through M22, Max column: Delete "-200" (two places) and substitute "0" (two places).

PAGE 39

FIGURE 16: Delete and substitute new figure 16 as printed on page 16 of this amendment.

PAGES 40 AND 41

TABLE XV, Test conditions N1 through N22, Min column: Delete "0" (two places) and substitute "-20" (two places).

PAGE 42

FIGURE 17: Delete and substitute new figure 17 as printed on page 17 of this amendment.

PAGE 43

TABLE XVI, Test conditions P1 through P16, Min column: Delete "0.0" (one place) and substitute "-0.4" (one place).

TABLE XVI, Test conditions P1 through P16, Max column: Delete "-0.4" (one place) and substitute "0.0" (one place).

PAGE 105

TABLE XLII, Test condition RR, Unit column: Delete "μF" and substitute "nF".

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Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0243)

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TABLE I. Module electrical requirements.

Test Cond	Parameter	Ref test table	Test type 1/-3/	Class I 4/ 25°C ± 5°C 4/ Min	Class I 4/ 25°C ± 5°C 4/ Max	Class I 4/ 0°C ± 5°C Min	Class I 4/ 0°C ± 5°C Max	Class I 4/ 60°C ± 5°C Min	Class I 4/ 60°C ± 5°C Max	Class II 4/ -55°C ± 5°C Min	Class II 4/ -55°C ± 5°C Max	Class II 4/ 85°C ± 5°C Min	Class II 4/ 85°C ± 5°C Max	Unit
A	Isolation	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	Initial	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
	End-of-life	IV	S	Module must pass all remaining tests										
B	VDD stress	V	S	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	V
C1-C22	Input clamp diode voltage, negative	VI	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D22	Input clamp diode voltage, positive	VII	X	See table VII										
E1-E7	Functional	VIII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F16	Output voltage, low level (IOL = 0)	IX	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
G1-G16	Output voltage, high level (IOH = 0)													

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
H1-H16	Output voltage, low level (VDD = 5 V)	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J1-J16	Output voltage, high level (VDD = 5 V)	XI	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
K1-K16	Output voltage, low level (VDD = 15 V)	XII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
L1-L16	Output voltage, high level (VDD = 15 V)	XIII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
M1-M22	Input current, low level	XIV	X	-200	0	---	---	-1000	0	---	---	-1000	0	nA
N1-N22	Input current, high level	XV	X	-20	200	---	---	-100	1000	---	---	-100	1000	nA
P1-P16	Output leakage current, low level (IOZL)	XVI	X	-0.4	0.0	---	---	-12.0	0.0	---	---	-12.0	0.0	µA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q16	Output leakage current, high level (IOZH)	XVII	X	0.0	0.4	---	---	0.0	12.0	---	---	0.0	12.0	µA
R1, R2	Power supply current	XVIII	X	0	80	0	80	0	1400	0	80	0	2400	µA
S1-S16	Transition time (tTLH), VDD = 5 V, CL = 50 pF	XIX	S	36	360	36	360	38	385	30	302	41	400	ns
S17-S32	Transition time (tTLH), VDD = 5 V, CL = 150pF	XIX	Q	84	840	84	840	91	916	74	743	100	960	ns
T1-T16	Transition time (tTHL), VDD = 5 V, CL = 50 pF	XX	S	20	200	20	200	21	213	17	171	22	218	ns
T17-T32	Transition time (tTHL), VDD = 5 V, CL = 150pF	XX	Q	44	440	44	440	47	478	35	354	52	500	ns

- See footnotes at end of table.

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Test Cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1-U16	Transition time (tTLH), VDD = 10 V, CL = 50pF	XXI	Q	18	180	18	180	19	191	15	156	20	198	ns
V1-V16	Transition time (tTHL), VDD = 10 V, CL = 50pF	XXII	Q	10	100	10	100	10	105	8	88	11	108	ns
W1-W16	Transition time (tTLH), VDD = 15 V, CL = 50pF	XXIII	Q	13	130	13	130	13	138	11	112	14	143	ns
X1-X16	Transition time (tTHL), VDD = 15 V, CL = 50pF	XXIV	Q	8	80	8	80	8	84	7	70	8	87	ns
Y1-Y16	Propagation delay time (CL to output), VDD = 5 V, CL = 50 pF, tPLH	XXV	S	60	600	60	600	61	611	57	576	62	618	ns

- See footnotes at end of table.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y17-Y32	Propagation delay time (CL to output), VDD = 5 V, CL = 50 pF, tPHL	XXV	S	60	600	60	600	61	611	57	576	62	618	ns
Z1-Z16	Propagation delay time (CL to output), VDD = 5 V, CL = 150 pF, tPLH	XXVI	Q	80	800	80	800	83	832	72	728	86	850	ns
Z17-Z32	Propagation delay time (CL to output), VDD = 5 V, CL = 150 pF, tPHL	XXVI	Q	80	800	80	800	83	832	72	728	86	850	ns
AA1-AA16	Propagation delay time (CL to output), VDD = 10 V, CL = 50 pF, tPLH	XXVII	Q	25	250	25	250	25	255	23	239	26	258	ns

- See footnotes at end of table.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
AA17-AA32	Propagation delay time (CL to output), VDD = 10 V, CL = 50 pF, tPHL	XXVII	Q	25	250	25	250	25	255	23	239	26	258	ns
BB1-BB16	Propagation delay time (CL to output), VDD = 15 V, CL = 50 pF, tPLH	XXVIII	Q	18	180	18	180	18	184	17	170	18	187	ns
BB17-BB32	Propagation delay time (CL to output), VDD = 15 V, CL = 50 pF, tPHL	XXVIII	Q	18	180	18	180	18	184	17	170	18	187	ns
CC1-CC16	Propagation delay time (Reset to output), VDD = 5 V, CL = 50 pF, tPHL	XXIX	S	60	600	60	600	61	611	57	576	62	618	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC17-CC32	Propagation delay time (Reset to output), VDD = 5 V, CL = 150 pF, tPHL	XXIX	Q	80	800	80	800	83	832	72	728	86	850	ns
DD1-DD16	Propagation delay time (Reset to output), VDD = 10 V, CL = 50 pF, tPHL	XXX	Q	25	250	25	250	25	255	23	239	26	258	ns
EE1-EE16	Propagation delay time (Reset to output), VDD = 15 V, CL = 50 pF, tPHL	XXXI	Q	18	180	18	180	18	184	17	170	18	187	ns
FF1-FF16	Disable delay time (tPLZ), VDD = 5 V, CL = 50 pF	XXXII	S	70	700	70	700	74	740	70	700	80	770	ns

- See footnotes at end of table.

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						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
FF17-FF32	Enable delay time (tPZL), VDD = 5 V, CL = 50 pF	XXXII	S	17	170	17	170	19	190	17	170	22	205	ns
FF33-FF48	Disable delay time (tPHZ), VDD = 5 V, CL = 50 pF	XXXII	S	0.15	1.50	0.15	1.50	0.16	1.60	0.15	1.50	0.17	1.65	µs
FF49-FF64	Enable delay time (tPZH), VDD = 5 V, CL = 50 pF	XXXII	S	14	140	14	140	16	160	14	140	19	175	ns
GG1-GG16	Disable delay time (tPLZ), VDD = 10 V, CL = 50 pF	XXXIII	Q	66	660	66	660	70	700	66	660	76	730	ns
GG17-GG32	Enable delay time (tPZL), VDD = 10 V, CL = 50 pF	XXXIII	Q	6	60	6	60	8	75	6	60	10	85	ns

- See footnotes at end of table.

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				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
GG33-GG48	Disable delay time (tPHZ), VDD = 10 V, CL = 50 pF	XXXIII	Q	0.15	1.50	0.15	1.50	0.16	1.60	0.15	1.50	0.17	1.65	µs
GG49-GG64	Enable delay time (tPZH), VDD = 10 V, CL = 50 pF	XXXIII	Q	6	60	6	60	8	75	6	60	10	85	ns
HH1-HH16	Disable delay time (tPLZ), VDD = 15 V, CL = 50 pF	XXXIV	Q	62	620	62	620	66	660	62	620	72	690	ns
HH17-HH32	Enable delay time (tPZL), VDD = 15 V, CL = 50 pF	XXXIV	Q	5	50	5	50	7	65	5	50	9	75	ns
HH33-HH48	Disable delay time (tPHZ), VDD = 15 V, CL = 50 pF	XXXIV	Q	0.14	1.40	0.14	1.40	0.15	1.50	0.14	1.40	0.16	1.55	µs

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
HH49-HH64	Enable delay time (tPZH), VDD = 15 V, CL = 50 pF	XXXIV	Q	5	45	5	45	6	60	5	45	8	70	ns
JJ1-JJ16	Hold time tH(H), VDD = 5 V, CL = 50 pF	XXXV	S	3	5	3	5	3	5	3	5	3	5	µs
JJ17-JJ32	Hold time tH(L), VDD = 5 V, CL = 50 pF	XXXV	S	3	5	3	5	3	5	3	5	3	5	µs
KK1-KK16	Setup time tS(H), VDD = 5 V, CL = 50 pF	XXXVI	S	3	5	3	5	3	5	3	5	3	5	µs
KK17-KK32	Setup time tS(L), VDD = 5 V, CL = 50 pF	XXXVI	S	3	5	3	5	3	5	3	5	3	5	µs
LL1-LL16	Hold time tH(H), VDD = 10 V, CL = 50 pF	XXXVII	Q	3	5	3	5	3	5	3	5	3	5	µs

- See footnotes at end of table.

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Test Cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
LL17-LL32	Hold time tH(L), VDD = 10 V, CL = 50 pF	XXXVII	Q	3	5	3	5	3	5	3	5	3	5	µs
MM1-MM16	setup time tS(H), VDD = 10 V, CL = 50 pF	XXXVIII	Q	3	5	3	5	3	5	3	5	3	5	µs
MM17-MM32	Setup time tS(L), VDD = 10 V, CL = 50 pF	XXXVIII	Q	3	5	3	5	3	5	3	5	3	5	µs
NN1-NN16	Hold time tH(H), VDD = 15 V, CL = 50 pF	XXXIX	Q	3	5	3	5	3	5	3	5	3	5	µs
NN17-NN32	Hold time tH(L), VDD = 15 V, CL = 50 pF	XXXIX	Q	3	5	3	5	3	5	3	5	3	5	µs
PP1-PP16	setup time tS(H), VDD = 15 V, CL = 50 pF	XL	Q	3	5	3	5	3	5	3	5	3	5	µs

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
PP17-PP32	Setup time tS(L), VDD = 15 V, CL = 50 pF	XL	Q	3	5	3	5	3	5	3	5	3	5	µs
QQ1-QQ16	Maximum shift frequency check	XLI	S	950	1150	950	1150	950	1150	950	1150	950	1150	ns
RR	Power supply filter capacitance	XLII	S	70	130	70	130	70	130	70	130	70	130	nF
SS1-SS8	Input capacitance	XLIII	Q	2	15	2	15	2	15	2	15	2	15	pF
SS9-SS16	Input capacitance	XLIII	Q	2	25	2	25	2	25	2	25	2	25	pF
SS17-SS22	Input capacitance	XLIII	Q	2	50	2	50	2	50	2	50	2	50	pF

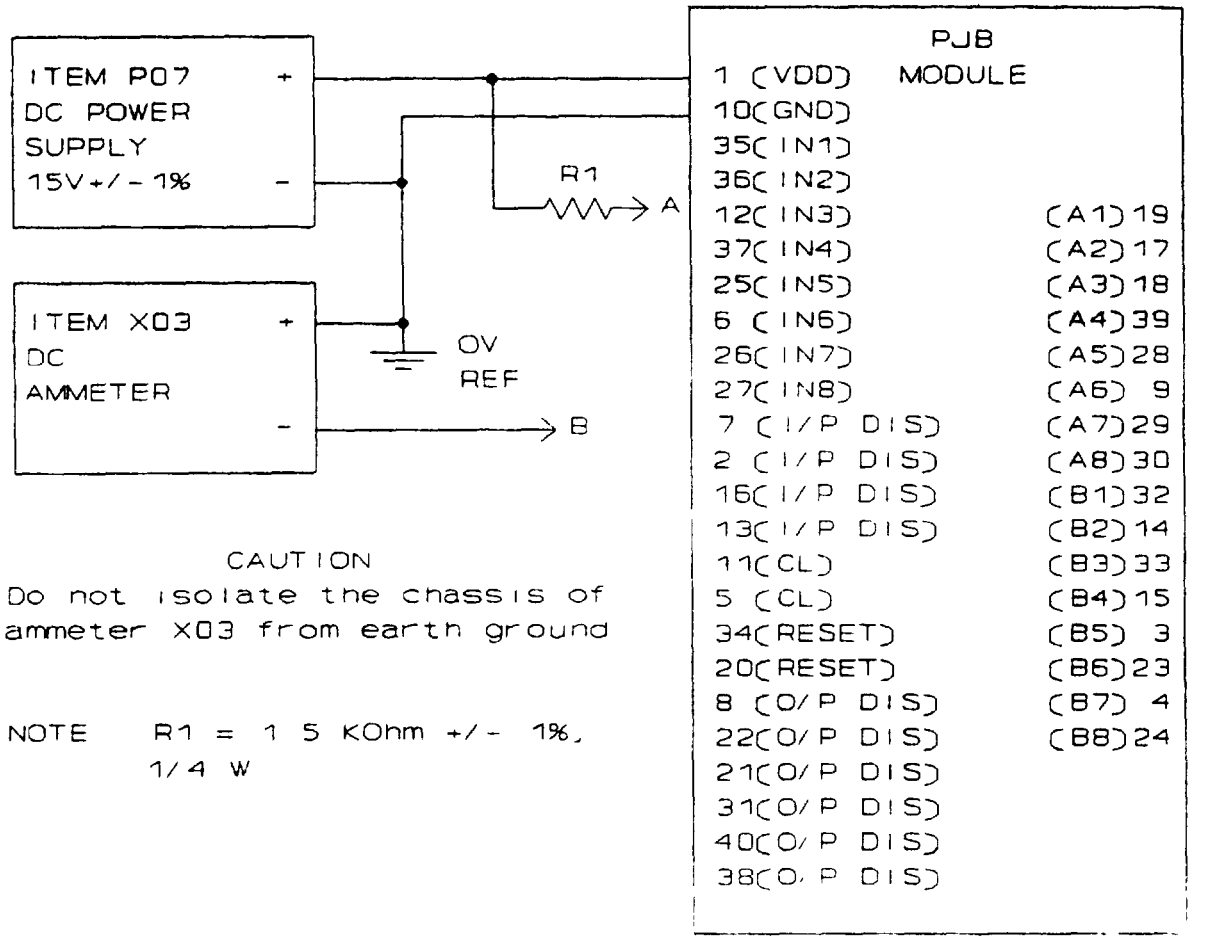
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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CAUTION
Do not isolate the chassis of ammeter X03 from earth ground

NOTE R1 = 15 KOhm +/- 1%, 1/4 W

FIGURE 16. Input current (low-level) test circuit for table XIV.

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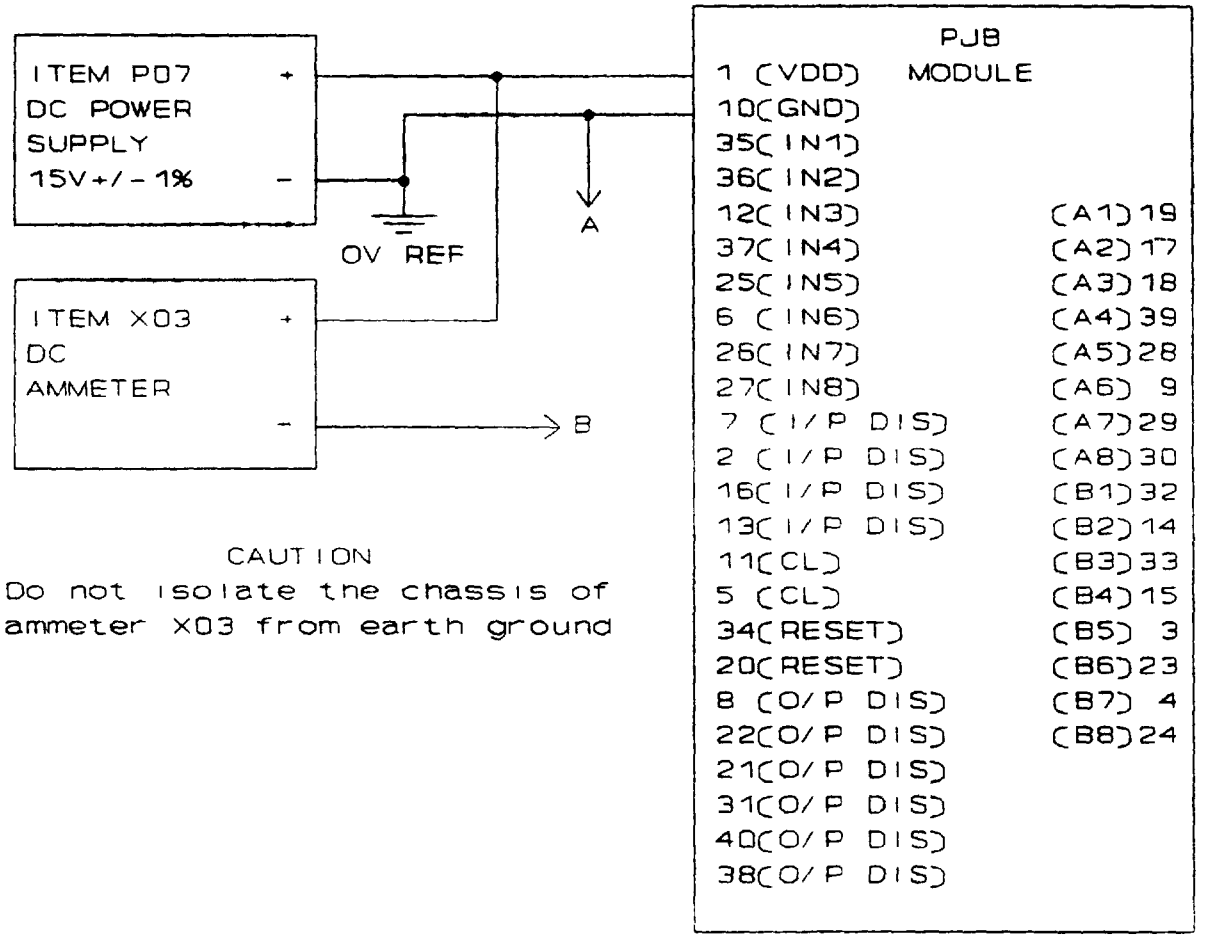


FIGURE 17. Input current (high-level) test circuit for table XV.

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